10th International Conference on Information Science, Signal Processing and their Applications (ISSPA 2010)

A SUBSPACE-BASED METHOD FOR SPEECH ENCRYPTION

Atef Mermoul and Adel Belouchrani

Electrical Engineering Department, Ecole Nationale Polytechnique, Algiers, Algeria.
atef.mermoul@enp.edu.dz, adel.belouchrani@enp.edu.dz

ABSTRACT

The use of signal processing techniques in cryptographic field has been an attractive approach in recent years. As an example, the intractability of the under-determined blind source separation (BSS) problem has been used for the proposal of BSS-based speech encryption. However, some weaknesses of this proposal from a cryptographic point of view have been recently published. In this paper, we propose a new encryption method that bypass these weaknesses. The proposed approach is based on the subspace concept together with the use of nonlinear functions and key signals. An interesting feature of the proposed technique is that only a part of the secret key parameters used during encryption is necessary for decryption. Furthermore, if no plain-text is fed to the encryption algorithm, the latter will provide no contents. Analysis results show that the proposed method enhances the security level of existing BSS-based speech encryption methods.

1. INTRODUCTION

Due to the rapid growth of digital communications and electronic data exchange, information security has become a crucial issue in industry, business, and administration. Modern cryptography provides essential techniques for securing information and protecting data. Among cryptographic techniques, Blind Source Separation (BSS) based methods have received recently some attention in speech and image encryption fields. Lin et al. proposed a series of different encryption schemes based on BSS [1, 2, 3]. In [1] and [2], the linear mixing model of blind source separation is used in image encryption. The transmitted images are covered with a noise image by specific mixing before encryption and then recovered through BSS after decryption [1]. A speech encryption algorithm which integrates a modified time domain scrambling scheme with an amplitude scrambling method was used to mask the speech signal with a random noise by specific mixing [3]. In [4], Lin et al. proposed a speech encryption scheme by taking advantage of the under-determined BSS problem. Their proposed approach constructs a mixing matrix for encrypting multiple segments simultaneously leading to enhancement of the security level of the previous proposed schemes. However, some weaknesses of the method proposed by Lin et al. from a cryptographic point of view are reported in [5]. The latter reference proves that the security of such BSS based approaches against some attacks is not sufficiently strong.

In this paper, a new approach based on the subspace concept is presented to bypass the above weaknesses.

2. PROBLEM FORMULATION

Let us first start by introducing blind source separation. The latter aims to recover a set of unknown mutually independent source signals from their observed mixtures without knowing the mixing coefficients [6]. Suppose that there exist M independent source signals and N observed mixtures of the source signals (usually $M \leq N$). The linear BSS mixing model is as follows:

$$\mathbf{x}(t) = \mathbf{H}\mathbf{s}(t) \tag{1}$$

where $\mathbf{s}(t) = [s_1(t), \cdots, s_M(t)]^T$, which is an $M \times 1$ column vector collecting the source signals, vector $\mathbf{x}(t)$ similarly collects the N observed (mixed) signals, and $\mathbf{H}$ is an $N \times M$ mixing matrix that contains the mixing coefficients.

The purpose of BSS is to find an $M \times N$ demixing matrix $\mathbf{W}$ such that the $M \times 1$ output vector $\mathbf{u}(t)$ verify

$$\mathbf{u}(t) = \mathbf{W}\mathbf{x}(t) = \mathbf{W}\mathbf{H}\mathbf{s}(t) = \mathbf{P}\mathbf{D}\mathbf{s}(t) \tag{2}$$

where $\mathbf{P}$ and $\mathbf{D}$ denote a permutation matrix and diagonal matrix, respectively. When $M \leq N$, the blind source separation and more specifically source recovery is possible. However, when $M > N$, BSS becomes generally impossible except under specific conditions [7, 8, 9]. This is referred to as the under-determined BSS problem.

In [4], the encryption procedure, including the use of key signals, has been represented as:

$$\mathbf{x}(t) = \mathbf{A}_s\mathbf{p}(t) + \mathbf{A}_k\mathbf{k}(t) \tag{3}$$

where

$$\begin{aligned} \mathbf{p}(t) &= [p_1(t), \cdots, p_M(t)]^T, \text{ and} \\ \mathbf{k}(t) &= [k_1(t), \cdots, k_M(t)]^T \end{aligned}$$

represent M input plain-signals and M key signals, respectively. $\mathbf{A}_s$ and $\mathbf{A}_k$ are $M \times M$ matrices, both of which elements are within $[-1, 1]$.

In the BSS-based encryption scheme [4], the key signals $k_1(t), \cdots, k_M(t)$ are as long as the plain-signals and have to be generated by a pseudo-random number generator (PRNG) with a secret seed, which serves as the secret key. The mixing matrices $\mathbf{A}_s$ and $\mathbf{A}_k$, being secret

978-1-4244-7167-6/10/$26.00 ©2010 IEEE

parameters, may be known by the receiver as secret keys and hence their estimation by a BSS approach at the receiver should not be necessary. Hence, the BSS approach is, in this case, worth to be used in a cryptanalysis process rather than in an encryption one.

In Li et al.'s paper [5], the security robustness of such BSS-based encryption scheme is evaluated. This cryptanalysis robustness study considers the Cipher-text-Only Attack approach in terms of the resistance level to Divide-and-Conquer (DAC) Attack and to Differential Attack, as well as the evaluating sensitivity to the mixing matrix, to the key signals and to the plain-text. In Known-Plain-text Attack approach, Li et al. [5] have evaluated the number of required plain-texts to yield at least M plain-text differentials. Taking advantage of these cryptanalysis considerations, a new approach for an encryption scheme based on the subspace concept is proposed in this paper.

3. PROPOSED METHOD

The main idea of the proposed approach consists of building an encryption scheme based on the subspace concept.

3.1. Encryption

The block diagram of the proposed encryption scheme is shown in Figure 1. The data are first fed to the segment splitter which consists of dividing the original speech into L segments:

$$\mathbf{p}(t) = [p_1(t), \cdots, p_M(t)]^T, t = 1, \cdots, L \qquad (4)$$

where M is the segment length. These segments are used in the encryption process (the subspace based encryption block) to obtain the following encrypted signal:

$$\mathbf{x}(t) = \mathbf{A}\mathbf{p}(t) + \beta \mathbf{P}_A^{\perp}\mathbf{B}[\mathbf{k}(t) \odot \mathbf{g}(\mathbf{p}(t))] \qquad (5)$$

where $\mathbf{A}$ is an $(M+1) \times M$ full rank key matrix. β is a factor that controls the signal to noise ratio [1], $\mathbf{g}(.)$ is a component-wise nonlinear function that verifies

$$\mathbf{g}(0) = 0. \qquad (6)$$

$\mathbf{k}(t)$ is a random $M \times 1$ key signal vector generated by any robust key signal generator and $\odot$ denotes the Hadamard operator. The matrix $\mathbf{B}$ is an arbitrary random $(M+1) \times M$ matrix and $\mathbf{P}_A^{\perp}$ is an orthogonal projector on the orthogonal subspace to the subspace spanned by the columns of the key matrix $\mathbf{A}$. The latter is refereed herein to as the key subspace. The projector $\mathbf{P}_A^{\perp}$ is given by

$$\mathbf{P}_A^{\perp} = \mathbf{I} - \mathbf{P}_A = \mathbf{I} - \mathbf{A}(\mathbf{A}^H\mathbf{A})^{-1}\mathbf{A}^H \qquad (7)$$

where $\mathbf{P}_A$ is the orthogonal projector on the key subspace, and $(.)^H$ and $\mathbf{I}$ denote the Hermitian operator and the identity matrix, respectively. For the purpose of robustness evaluation, we use in the sequel the following component-wise nonlinear function:

$$g(v) = \frac{v}{\sqrt{1+v^2}} \qquad (8)$$

that verifies condition (6). Note that in the absence of plain-text (i.e. $\mathbf{p}(t) = 0$), the crytosystem will provide no contents (i.e. $\mathbf{x}(t) = 0$). This important property allows to ban the use of any signal processing based crytoanalysis technique in absence of plain-text.

The originality of this contribution consists of the judicious choice of the second term of equation (5). This term is made as complicated as possible to not be recovered or attacked by the cryptanalysis techniques presented in reference [5]. Note also that this additive term, in addition to be nonlinear and correlated with the first term, vanishes if no plain-text is provided.

3.2. Decryption

On the receiver hand, the encrypted data is first projected on the key subspace; this is done by the following operation:

$$\mathbf{x}_p(t) = \mathbf{P}_A\mathbf{x}(t) \qquad (9)$$

where $\mathbf{x}_p(t)$ is the obtained projected data. Since the projectors $\mathbf{P}_A$ and $\mathbf{P}_A^{\perp}$ are orthogonal (i.e. $\mathbf{P}_A\mathbf{P}_A^{\perp} = \mathbf{0}$), the above projection leads to the following result

$$\mathbf{x}_p(t) = \mathbf{A}\mathbf{p}(t) \qquad (10)$$

and the original plain-text (the decrypted signal) is obtained by using the key matrix $\mathbf{A}$:

$$\mathbf{p}(t) = \mathbf{A}^{\sharp}\mathbf{x}_p(t) \qquad (11)$$

where $(.)^{\sharp}$ denotes the pseudo-inverse operator.

Note that in the above recovery procedure, one does not need to know the key signals $\mathbf{k}(t)$ neither the matrix $\mathbf{B}$. Figure 2 shows the results obtained from the use of the proposed method when applied on a speech file containing a record of the first author daughter singing "Let us laugh together". The original signal was sampled at a rate of 22.05 KHz and the number of bits per sample used to encode the data in the file was 16 bits/sample. Figure 2-(a) shows the original speech signal and Figure 2-(b) shows the key signal used in the encryption process. This key signal[2] is generated using a pseudo-random key generator. Of course, other types of key signal generators can be used. This issue is out of the scope of this paper. Figure 2-(c) shows the signal encrypted according to equation (5) with a segment length equal to $M = 4$. Note that the encrypted signal has more samples than the original one, actually L samples more where L is the segment number. This sample excess comes from the fact that the dimension of the key matrix $\mathbf{A}$ is $(M+1) \times M$. After decryption with the proposed subspace method, the recovered signal is shown in Figure 2-(d).

[1] This factor (β) should be chosen as large as possible in order to provide very low SNR.

[2] The splitting of this key signal according to the same format used for the speech signal (see equation (4)) leads to the key signal vector $\mathbf{k}(t)$.

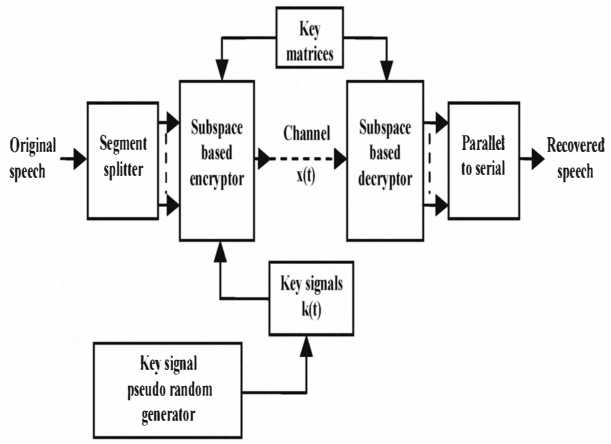


Fig. 1. Block diagram of the proposed subspace-based speech Encryption.

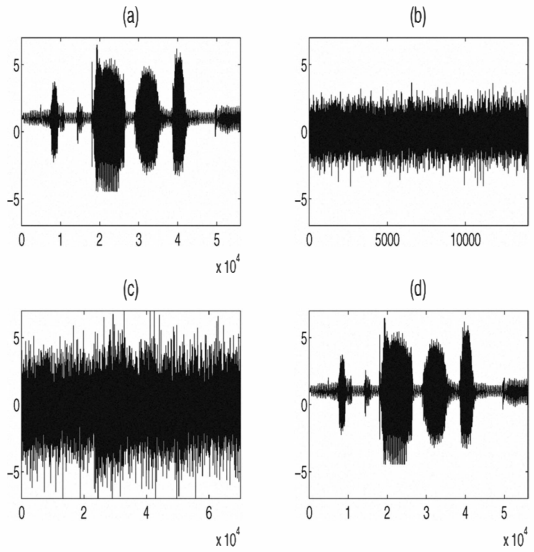


Fig. 2. Sample run of the proposed subspace-based encryption method: (a) Original Signal, (b) Key signal, (c) Encrypted signal, (d) Decrypted signal.

4. EVALUATION

In this section, the cryptographic robustness of the proposed subspace-based encryption scheme is evaluate according to Cipher-text-only attack and to Differential attack .

4.1. Cipher-text-only attack: Sensitivity to plain-text

A cryptographic property required by a robust cryptosystem is that the encryption must be very sensitive to plain-text, i.e., the cipher-texts of two plain-texts with a slight difference should be much different [5]. This property matches well with the proposed subspace-based encryption scheme. Using the same key and given two plain-texts $p^{(1)}(t)$ and $p^{(2)}(t)$ with a very slight difference, the obtained result presents a significant sensitivity to this slight difference. Figure 3 shows an example of this property where $p^{(2)}(t)$ has been set equal to $p^{(1)}(t)+\epsilon$. Figure 3-(a) shows the difference between two cipher-texts obtained from the encryption of two slightly different plain-texts ($\epsilon = 0.1$) by using the same key. Other values of ϵ have been tested to evaluate the sensitivity to plain-texts. Figure 3-(b) shows this comparison for $\epsilon = 0.01$. It has been tested that no one could detect the slight difference between the two plain-texts (speech signals). In other words, for a person, these two speech signals are actually the same. However, their encrypted versions are too different.

4.2. Differential attack

A differential attack, as described by Li et al. [5], is based on the assumption that two identical key signals are used to encrypt at least two plain-texts. However, the key space of a good pseudo random number generator, which generates pseudo random sequences having statistical properties similar to those of random sequences, should be large enough to prevent the occurrence of two identical key signals. Moreover, even when we assume that two identical key signals have been used to encrypt two plain-texts, the differential attack can not be realized.

Let us assume that two plain-texts $\mathbf{p}^{(1)}(t)$ and $\mathbf{p}^{(2)}(t)$ are encrypted using the same key parameters $(\mathbf{P}_A, \mathbf{k}(t))$. From equation (5), one has:

$$\mathbf{x}^{(1)}(t) = \mathbf{A}\mathbf{p}^{(1)}(t) + \beta\mathbf{P}_A^{\perp}\mathbf{B}[\mathbf{k}(t) \odot \mathbf{g}(\mathbf{p}^{(1)}(t))] \quad (12)$$

and

$$\mathbf{x}^{(2)}(t) = \mathbf{A}\mathbf{p}^{(2)}(t) + \beta\mathbf{P}_A^{\perp}\mathbf{B}[\mathbf{k}(t) \odot \mathbf{g}(\mathbf{p}^{(2)}(t))] \quad (13)$$

Combining equations (12) and (13), leads to

$$\Delta_x(t) = \mathbf{A}\Delta_p(t)+\beta\mathbf{P}_A^{\perp}\mathbf{B}[\mathbf{k}(t)\odot[\mathbf{g}(\mathbf{p}^{(1)}(t))-\mathbf{g}(\mathbf{p}^{(2)}(t))]] \quad (14)$$

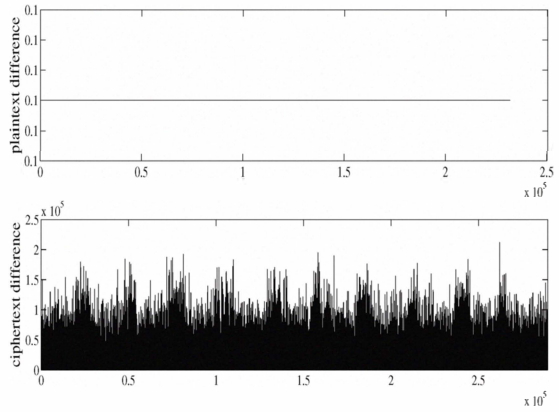
where $\Delta_x(t)$ is the cipher-text differential described as:

$$\Delta_x(t) = \mathbf{x}^{(1)}(t) - \mathbf{x}^{(2)}(t) \quad (15)$$

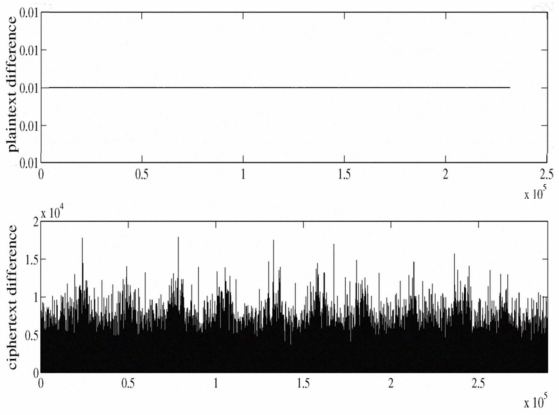
and $\Delta_p(t)$ is the plain-text differential described as:

$$\Delta_p(t) = \mathbf{p}^{(1)}(t) - \mathbf{p}^{(2)}(t) \quad (16)$$

Note that even if the same key is used to obtain the cipher-texts $\mathbf{x}^{(1)}(t)$ and $\mathbf{x}^{(2)}(t)$, the additive subspace perturbation term is still present in equation (14). Moreover, the plain-text differential $\Delta_p(t)$ can not be computed because of the permanent presence of the terms $\mathbf{p}^{(1)}(t)$ and $\mathbf{p}^{(2)}(t)$ in equation (14). This is due, as mentioned in Section 3, to the existing correlation between the additive subspace perturbation term and the plain-text term of the proposed encryption equation (5).



(a)



(b)

Fig. 3. Sensitivity to plain-texts: plain-text difference and their corresponding cipher-text difference, (a) $\epsilon = 0.1$, (b) $\epsilon = 0.01$.

5. CONCLUSION

In this paper, a new speech encryption method based on the subspace concept is proposed. Our main contribution consists of the judicious choice of the second term of the encryption equation (5). This term is made as complicated as possible to not be recovered or attacked by the cryptanalysis techniques presented in reference [5]. In addition of being nonlinear and correlated with the plain-text, this additive term vanishes if no plain-text is provided. This choice is motivated by the fact that in statistical signal processing such problem has to our no known solution in the open litterature. We have also proved that the proposed subspace-based cryptosystem presents a high degree of immunity against cipher-text-only and differential attacks. These attacks constitute vulnerabilities [5] of the BSS-based encryption scheme proposed in [4]. Another interesting characteristic of the proposed subspace-based encryption is that the key parameters, except the key matrix $\mathbf{A}$, used during encryption are not necessary for decryption. It remains that further evaluations of the robustness of this subspace-based encryption method to other types of cryptanalysis attacks should be conducted.

6. REFERENCES

[1] Q.-H. Lin, and F.-L. Yin "Blind source separation applied to image cryptosystems with dual encryption ," in Electronics Letters, vol. 38, no. 19, pp. 1092-1094, September 2002.

[2] Q.-H. Lin, F.-L. Yin, and Y.-R. Zheng "Secure image communication using blind source separation," in IEEE 6th CAS Symp. on Emerging Technologies: Mobile and Wireless CO. Shanghai, China, May 31-June 2,2004.

[3] Q.-H. Lin, F.-L. Yin, T.-M. Mei, and H.-L. Liang, "A speech encryption algorithm based on blind source separation," in in Proc. Int. Conf. Commun., Circuits Syst.: Signal Process., Circuits Syst., 2004, vol. II, pp. 1013-1017.

[4] Q.-H. Lin, F.-L. Yin, T.-M. Mei, and H. Liang, "A blind source separation based method for speech encryption," in IEEE Trans. Circuits Syst. I, vol. 53, no. 6, pp. 1320-1328, June 2006.

[5] S. Li, C. Li, K.-T. Lo, and G. Chen, "Cryptanalysis of an encryption scheme based on blind source separation," in IEEE Trans. Circuits Syst. I, vol. 55, no. 4, pp.1055-1063, 2008.

[6] A. Belouchrani, K. A. Meraim, J.-F. Cardoso, and E. Moulines, "A blind source separation technique using second-order statistics," in IEEE Trans. Signal Processing, vol. 45, no. 2, pp. 434-444, February 1997.

[7] A. Belouchrani and J.F. Cardoso, "A maximum likelihood source separation for discrete sources", in Proc. EUSIPCO, Vol. 2, pp. 768-771, September 1994.

[8] P. Comon, "Blind identification and source separation in 2 3

[9] D. Aissa EL Bey, K. Abed-Meraim, A. Belouchrani and Y. Grenier, "Underdetermined Blind Separation of Nondisjoint Sources in the Time-Frequency Domain", IEEE Trans. on Signal processing, Vol. 55, Issue 3, pp. 897-907, Mar. 2007.